

BUZAU, Romania

WMO: 153500

Lat: 45.1328N

Lon: 26.8517E

Elev: 323

StdP: 14.53

Time Zone: 2.00 (EUE)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	10.3	14.8	3.4	6.6	13.3	8.3	8.5	17.5	36.4	29.1	26.6	25.5	7.7	50	0.493

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	19.5	92.9	70.3	89.9	69.4	87.2	68.8	74.6	86.0	72.5	83.9	70.8	82.3	7.1	190

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
71.0	115.5	80.8	68.8	107.1	78.2	67.0	100.5	76.4	38.2	86.8	36.4	84.0	34.8	82.2	84.7

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 23.3	20.0	17.1	DB	2.6	97.4	6.0	3.2	-1.8	99.8	-5.3	101.7	-8.7	103.5	-13.1	105.8	(4)
(5)			WB	2.0	76.6	5.9	3.1	-2.3	78.8	-5.8	80.7	-9.1	82.4	-13.4	84.7	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	53.7	30.3	34.9	43.6	54.0	64.0	71.0	74.8	74.7	65.3	54.0	43.6	33.2		
	DBStd	17.24	8.47	9.27	8.14	6.77	6.18	5.59	4.99	5.51	6.42	7.38	7.65	8.14		
	HDD50	2086	612	424	228	37	1	0	0	0	0	45	219	520		
	HDD65	5078	1077	843	664	335	93	14	1	3	75	346	642	985		
	CDD50	3441	0	2	28	156	436	630	770	764	458	170	27	0		
	CDD65	958	0	0	0	4	63	194	306	302	83	6	0	0		
	CDH74	8803	0	0	2	62	583	1715	2863	2865	655	58	0	0		
	CDH80	3303	0	0	0	9	127	580	1177	1225	179	6	0	0		
(14) Wind	WSAvg	7.1	7.3	7.9	8.8	8.0	7.2	6.5	6.5	6.4	6.6	6.4	6.7	7.1		
(15) Precipitation	PrecAvg	21.1	1.0	0.7	1.1	1.7	2.6	3.0	2.6	2.0	2.0	1.8	1.3	1.3		
	PrecMax	30.0	2.4	2.2	2.4	4.3	7.7	4.6	6.7	5.1	4.6	5.2	3.0	3.5		
	PrecMin	13.5	0.1	0.1	0.0	0.7	0.7	0.9	0.3	0.4	0.2	0.1	0.0	0.0		
	PrecStd	4.4	0.6	0.6	0.6	0.8	1.6	1.0	1.7	1.3	1.2	1.2	0.9	0.9		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	55.7	62.5	71.7	80.9	87.7	93.0	97.2	97.9	89.8	80.2	68.9	56.3		
		MCWB	45.7	51.6	53.6	61.3	66.8	70.9	69.4	70.0	66.7	63.1	57.6	49.1		
	2%	DB	49.3	57.2	66.4	74.6	83.4	89.0	92.9	93.3	84.9	74.7	63.0	50.9		
		MCWB	42.5	48.1	51.9	58.8	64.9	70.0	70.1	70.6	65.7	61.4	54.9	46.2		
	5%	DB	44.4	52.0	61.6	70.3	80.2	86.2	89.7	90.1	81.2	70.0	58.2	46.9		
		MCWB	40.7	45.1	49.6	56.4	63.4	69.0	69.9	68.9	63.9	58.9	52.1	43.0		
	10%	DB	41.2	47.2	57.1	66.5	77.0	83.2	86.7	87.0	77.5	65.6	54.5	43.6		
		MCWB	38.4	42.2	47.2	54.3	61.9	67.6	69.4	68.1	62.7	56.8	50.3	40.6		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	47.1	53.1	56.8	63.0	70.8	75.1	77.2	78.8	72.5	66.2	58.9	50.8		
		MCDB	53.8	59.9	67.1	77.3	84.5	86.8	86.9	89.2	83.5	75.3	66.6	54.5		
	2%	WB	43.6	48.9	52.9	60.0	67.4	72.5	74.4	74.8	68.3	62.8	56.0	46.8		
		MCDB	47.5	55.6	63.6	72.0	78.6	84.7	85.7	86.2	77.9	70.4	62.1	50.1		
	5%	WB	41.4	45.7	50.5	57.9	65.3	70.8	72.6	71.9	65.8	60.0	53.4	43.7		
		MCDB	44.3	50.8	60.5	67.8	76.5	82.5	83.8	82.9	76.8	68.2	57.0	46.2		
	10%	WB	38.6	42.9	47.9	55.8	63.3	69.1	71.0	69.7	64.1	57.8	50.8	40.9		
		MCDB	41.0	47.0	55.5	64.2	74.0	80.1	82.2	82.1	74.2	64.3	54.1	43.2		
(35) Mean Daily Temperature Range	5% DB	MDBR	11.8	13.8	17.0	18.7	19.8	19.4	19.5	20.1	18.6	16.4	13.3	11.5		
		MCDBR	17.2	22.7	26.5	25.3	24.5	23.5	24.1	24.2	23.9	23.2	20.9	17.3		
	5% WB	MCWBR	12.1	14.8	14.8	12.7	10.7	9.3	8.4	8.2	9.3	11.5	13.1	12.3		
		MCWBR	16.1	21.2	24.2	22.4	22.0	21.4	20.4	21.2	21.2	21.2	18.3	15.8		
(40) Clear-Sky Solar Irradiance	taub		0.333	0.363	0.398	0.455	0.430	0.433	0.447	0.461	0.422	0.385	0.358	0.335		
	taud		2.332	2.248	2.205	2.061	2.216	2.241	2.207	2.156	2.230	2.315	2.345	2.343		
	Ebn at Noon		241	254	262	255	267	266	260	251	251	244	229	227		
	Edh at Noon		25	33	39	49	44	43	44	45	38	30	25	23		
(44) All-Sky Solar Radiation	RadAvg		417	679	1050	1444	1817	1938	1971	1774	1288	835	457	350		
	RadStd		56	88	110	149	140	140	138	112	107	95	49	46		

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		+0.77	N/A	N/A	N/A	N/A	-1.26	N/A	N/A	N/A	N/A
(47) Regional (58 neighbors)		+0.90	N/A	N/A	N/A	N/A	N/A	N/A	-257	+139	+74

Nomenclature: See separate page